

Biotechnology: A Laboratory Skills Course

Second Edition

Correlation Table for the Biotechnology Aptitude and Competency Exam

Biotechnology: A Laboratory Skills Course, Second Edition provides background content and activities that support student learning in the biotechnology topics and skills assessed on the Biotechnology Aptitude and Competency Exam (BACE) from Biotility. The following correlation table describes how and where each topic is addressed in the student text.

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Technical Skills and Applications

Topic	Textbook Content	Lab Activity
Discuss current techniques used in biotechnology, and their applications	<i>Throughout</i>	<i>Throughout</i>
Describe the proper use of microscopes	3.1 Microbiology and Cell Biology 3.2 Bacteria 3.8 Eukaryotic Cell Culture 8.3 Immunoassays	Activity 3.C Microbes and Health: An Illustration of Koch's Postulates Activity 3.D Gram Staining Activity 3.E Quantifying Bacterial Numbers Activity 3.F Staining Eukaryotic Cells
Discuss cell staining, and distinguish between Gram positive/negative cells	3.2 Bacteria 3.5 Microbiological Techniques	Activity 3.D Gram Staining
Describe the process of culturing microorganisms and tissues using aseptic technique	2.3 Laboratory Equipment 3.5 Microbiological Techniques 3.8 Eukaryotic Cell Culture [How To] Use Aseptic Technique to Transfer Bacteria	Activity 3.B Disk Diffusion Test (Modified Kirby-Bauer Test) Activity 3.C Microbes and Health: An Illustration of Koch's Postulates Activity 3.E Quantifying Bacterial Numbers Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography
Discuss the differences between sterilization, decontamination, and disinfection	2.3 Laboratory Equipment 3.5 Microbiological Techniques 3.8 Eukaryotic Cell Culture [How To] Use Aseptic Technique to Transfer Bacteria	
Understand the principle by which a pH meter works	2.3 Laboratory Equipment	Activity 2.D Making Solutions Activity 2.E Titration
Explain the principles of spectrophotometry	3.5 Microbiological Techniques 5.3 Plasmid Purification and Quantitation 7.5 Methods of Protein Analysis	Activity 5.D DNA Quantitation Activity 7.A Protein Quantitation Using the Bradford Assay Activity 7.E Biofuel Enzyme Assay Activity 8.E Quantitative ELISA



Understand the basic principle of Beer's Law	3.5 Microbiological Techniques 5.3 Plasmid Purification and Quantitation 7.5 Methods of Protein Analysis 9.2 Student Research Projects	Activity 5.D DNA Quantitation Activity 7.A Protein Quantitation Using the Bradford Assay Activity 7.E Biofuel Enzyme Assay Activity 8.E Quantitative ELISA
Discuss methods of chromosomal and plasmid DNA isolation, purification, and quantification	5.3 Plasmid Purification and Quantitation	Activity 2.A DNA Extraction and Precipitation Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation
Discuss gel electrophoresis techniques, including agarose and polyacrylamide gel electrophoresis (PAGE)	4.4 DNA Analysis Techniques 7.5 Methods of Protein Analysis [How To] Set Up a Vertical Electrophoresis System	Activity 4.C Dye Electrophoresis Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.D DNA Quantitation Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding Activity 7.D SDS-PAGE of Fish Muscle Activity 8.D Western Blotting
Understand how restriction enzymes are used	4.3 Recombinant DNA Technology [How To] Set Up a Restriction Digest	Activity 4.A Restriction Site Prediction Using NEBcutter Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 4.F Plasmid Mapping
Describe recombinant DNA and cloning techniques	4.3 Recombinant DNA Technology [How To] Set Up a Restriction Digest [How To] Transform <i>E.coli</i> Using Electroporation	Activity 4.A Restriction Site Prediction Using NEBcutter Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 4.F Plasmid Mapping Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 5.C Purification of S3 and pGLO Plasmids
Describe the mechanism of polymerase chain reaction (PCR)	Chapter 6 The Polymerase Chain Reaction [How To] Program a Thermal Cycler	Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding
Discuss protein expression in model organisms	1.2 Who Uses Biotechnology? 3.2 Bacteria 3.7 Uses of Eukaryotic Cells in Biotechnology 4.3 Recombinant DNA Technology 5.1 History of Bacterial Transformation and Plasmids 7.1 Protein Synthesis 7.4 Proteins in Biotechnology 7.6 Protein Production in Industry [Biotech in the Real World] Biotech on the Pharm [Biotech in the Real World] Vaccines in Bananas	Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 6.B GMO Detection by PCR Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography

Discuss methods of molecule/protein isolation, purification, and quantification	7.5 Methods of Protein Analysis [How To] Set Up a Vertical Electrophoresis System	Activity 7.A Protein Quantitation Using the Bradford Assay Activity 7.B Size Exclusion Chromatography Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.D SDS-PAGE of Fish Muscle Activity 7.E Biofuel Enzyme Assay
Understand western blotting, enzyme-linked immunosorbent assay (ELISA), and other immunoassays	8.3 Immunoassays [How To] Select an Antibody	Activity 8.A Ouchterlony Double Immunodiffusion Assay Activity 8.B Serum Antibody Detection by ELISA Activity 8.C Quantitative ELISA Activity 8.D Western Blotting
Understand molecular diagnostics techniques and their applications	1.2 Who Uses Biotechnology? 4.4 DNA Analysis Techniques 4.5 Restriction Enzymes as Forensic Tools 6.2 What Is PCR? 6.4 Types of PCR 6.6 Techniques Based on PCR 6.7 Real-World Applications of PCR 7.5 Methods of Protein Analysis 8.2 Antibodies as Tools [Careers in Biotech] Joshua Moore [Biotech in the Real World] Fighting Crime with DNA	Activity 4.E Forensic DNA Fingerprinting Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 8.A Ouchterlony Double Immunodiffusion Assay Activity 8.B Serum Antibody Detection by ELISA Activity 8.C Quantitative ELISA Activity 8.D Western Blotting
Discuss assay development and validation	9.2 Student Research Projects	Activity 7.E Biofuel Enzyme Assay
Understand the principles and applications of flow cytometry and cell sorting	8.3 Immunoassays	
Discuss the principles of centrifugation and its applications	5.3 Plasmid Purification and Quantitation [How To] Use a Microcentrifuge	Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.E Biofuel Enzyme Assay
Understand the basics of chromatography techniques	1.1 What Is Biotechnology? Chapter 2 Extension Activities 5.3 Plasmid Purification and Quantitation 7.5 Methods of Protein Analysis	Activity 2.C Kool-Aid Column Chromatography Activity 5.C Purification of S3 and pGLO Plasmids Activity 7.B Size Exclusion Chromatography Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography
Discuss the transformation or transfection of model organisms	5.2 Transforming Cells [How To] Transform <i>E. coli</i> Using Electroporation	Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid

Understand the principles of lab automation and robotics in biotechnology	[Biotech in the Real World] Fighting Crime with DNA [Careers In Biotech] Timothy Balmorez 8.3 Immunoassays	
Explain the importance of equipment calibration for accurate measurements	1.5 Industry Practices 2.3 Laboratory Equipment [How To] Use an Adjustable-Volume Micropipet	Activity 2.B Pipetting Activity 2.C Kool-Aid Column Chromatography Activity 2.D Making Solutions Activity 2.E Titration
Discuss how validation ensures a process or equipment operates as intended	1.3 The Biotechnology Industry 1.5 Industry Practices 9.2 Student Research Projects	
Discuss the principles and applications of laminar flow in a cleanroom or manufacturing environment	2.1 Laboratory Safety 3.5 Microbiological Techniques	

Safety and Workplace Culture

Topic	Textbook Content	Lab Activity
Discuss ethics and bioethics in the workplace and society	<i>Throughout (vignettes)</i>	
Describe appropriate workplace behaviors	1.5 Industry Practices 1.7 Careers in Biotechnology 2.1 Laboratory Safety 2.2 Laboratory Notebooks 2.3 Laboratory Equipment 2.4 Numerical Data	
Identify proper workplace safety behaviors	1.5 Industry Practices 1.7 Careers in Biotechnology 2.1 Laboratory Safety 2.2 Laboratory Notebooks 2.3 Laboratory Equipment 2.4 Numerical Data	
Identify safety symbols	2.1 Laboratory Safety 2.3 Laboratory Equipment	
Identify and explain proper use of safety equipment	2.1 Laboratory Safety	
Identify and properly use personal protective equipment (PPE)	2.1 Laboratory Safety	<i>Throughout</i>
Exercise proper safety protocols	1.3 The Biotechnology Industry 2.1 Laboratory Safety 2.3 Laboratory Equipment	<i>Throughout</i>

Describe proper handling of biological and hazardous waste	1.3 The Biotechnology Industry 2.1 Laboratory Safety 9.2 Student Research Projects [Bioethics] Waste Disposal	Activity 3.A Making Microbiology Media Activity 3.B Disk Diffusion Test (Modified Kirby-Bauer Test) Activity 3.C Microbes and Health: An Illustration of Koch's Postulates Activity 3.D Gram Staining Activity 3.E Quantifying Bacterial Numbers Activity 3.F Staining Eukaryotic Cells Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 5.C Purification of S3 and pGLO Plasmids Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.D SDS-PAGE of Fish Muscle
Explain the importance of posting and complying to signage	2.1 Laboratory Safety	
Describe the procedures for safe handling and storage of chemicals	2.1 Laboratory Safety 9.2 Student Research Projects	
Derive information from safety data sheets (SDS)	2.1 Laboratory Safety	
Discuss key Occupational Safety and Health Administration (OSHA) regulations applicable to biotechnology workplaces	1.4 Governmental Regulation of Biotechnology 2.1 Laboratory Safety	
Describe proper lockout/tagout procedures for machinery and equipment	–	–
Describe emergency response procedures for fires, chemical spills, or other incidents	2.1 Laboratory Safety 2.3 Laboratory Equipment	
Discuss the process for reporting safety concerns and hazards, including near misses	–	–
Discuss the importance of proactive safety communication	–	–

Understand the importance of comprehensive training and safety training in biotechnology workplaces

1.5 Industry Practices
2.1 Laboratory Safety
2.2 Laboratory Notebooks

Properly label items, including solutions, buffers, Petri plates, samples, and products

2.1 Laboratory Safety
2.3 Laboratory Equipment

Biochemistry and Molecular Biology

Topic	Textbook Content	Lab Activity
Understand the chemistry of molecules and macromolecules	<i>Primarily covered in chemistry and general biology courses. DNA structure and function are discussed in Chapter 4. Protein structure and function are explained in Chapter 7.</i>	Activity 2.A DNA Extraction and Precipitation Activity 2.C Kool-Aid Column Chromatography Activity 2.E Titration Activity 3.A Making Microbiology Media Activity 3.D Gram Staining Activity 4.C Dye Electrophoresis Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation Activity 7.A Protein Quantitation Using the Bradford Assay Activity 7.B Size Exclusion Chromatography Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.D SDS-PAGE of Fish Muscle Activity 7.E Biofuel Enzyme Assay
Describe DNA structure and function	4.1 Molecular Biology 4.2 DNA Structure 5.1 History of Bacterial Transformation and Plasmids 6.2 What Is PCR? 7.1 Protein Synthesis	Activity 4.A Restriction Site Prediction Using NEBcutter Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 4.F Plasmid Mapping Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding
Describe transcription	4.1 Molecular Biology 5.1 History of Bacterial Transformation and Plasmids 7.1 Protein Synthesis	Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid

Describe translation and gene expression	1.2 Who Uses Biotechnology? 3.7 Uses of Eukaryotic Cells in Biotechnology 4.1 Molecular Biology 4.3 Recombinant DNA Technology 5.1 History of Bacterial Transformation and Plasmids 7.1 Protein Synthesis [Biotech in the Real World] Biotech on the Pharm	Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid
Describe protein structure and function	Chapter 7 Protein Structure and Analysis Chapter 8 Immunological Applications	Activity 7.B Size Exclusion Chromatography Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.D SDS-PAGE of Fish Muscle Activity 7.E Biofuel Enzyme Assay Activity 8.A Ouchterlony Double Immunodiffusion Assay Activity 8.B Serum Antibody Detection by ELISA Activity 8.C Quantitative ELISA Activity 8.D Western Blotting
Explain how enzymes function and affect reaction rates	7.3 Proteins in Biology	Activity 7.E Biofuel Enzyme Assay
Understand cell settling and centrifugation techniques	5.3 Plasmid Purification and Quantitation [How To] Use a Microcentrifuge	Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.E Biofuel Enzyme Assay
Discuss proper techniques for mixing solutions and preparing reagents	2.5 Preparing Solutions	Activity 2.D Making Solutions Activity 2.E Titration
Understand the general physiology of cells	3.1 Microbiology and Cell Biology 3.2 Bacteria 3.6 Eukaryotic Cells	
Explain the interaction between cells, and between cells and their environment	3.3 Uses of Bacteria in Biotechnology 3.6 Eukaryotic Cells 5.1 History of Bacterial Transformation and Plasmids 7.1 Protein Synthesis 7.2 Protein Structure	Activity 5.B Bacterial Transformation with pGLO Plasmid
Understand the genetics of model organisms	1.3 The Biotechnology Industry 1.4 Governmental Regulation of Biotechnology 3.1 Microbiology and Cell Biology [Biotech in the Real World] Bio-Rad: Then and Now	Activity 3.C Microbes and Health: An Illustration of Koch's Postulates Activity 5.B Bacterial Transformation with pGLO Plasmid

Discuss monoclonal antibody (mAb) production and applications	8.2 Antibodies as Tools	
Understand the role of mRNA in cellular function and therapeutics	3.6 Eukaryotic Cells 4.1 Molecular Biology 5.1 History of Bacterial Transformation and Plasmids 6.2 What Is PCR? 6.4 Types of PCR 6.6 Techniques Based on PCR 7.1 Protein Synthesis	Activity 5.B Bacterial Transformation with pGLO Plasmid
Discuss immunotherapy types and applications, such as chimeric antigen receptor (CAR) T cells and mAb	1.2 Who Uses Biotechnology? [Biotech in the Real World] CAR T Cell Therapy — A New Horizon in Cancer Treatment	

Regulation and Quality

Topic	Textbook Content	Lab Activity
Discuss regulatory agencies governing the manufacture and distribution of biotechnology-derived products	1.4 Governmental Regulation of Biotechnology	
Outline the role of various departments in a company, including Research and Development, Quality Assurance, Quality Control, and Manufacturing	1.6 Biotechnology Company Structure	
Understand the purpose of Good Laboratory Practices (GLPs) in product development	1.5 Industry Practices	
Understand the purpose and importance of current Good Manufacturing Practices (cGMPs)	1.5 Industry Practices	
Follow practices associated with regulatory compliance	1.5 Industry Practices	

Discuss key components of Good Documentation Practices (GDocPs), including contemporaneous documentation, controlled documents, and approvals

2.2 Laboratory Notebooks
[How To] Write an SOP

Identify the proper use and essential elements of cGMP documents, such as standard operating procedures (SOPs), batch records, logs, and deviation reports

1.5 Industry Practices

Activity 2.F Writing a Standard Operating Procedure

Explain the research and development U.S. Food and Drug Administration (FDA) process for regulated products such as biopharmaceuticals and medical devices

1.4 Governmental Regulation of Biotechnology

Explain the FDA approval process for regulated products such as biopharmaceuticals and medical devices

1.4 Governmental Regulation of Biotechnology

Outline the manufacturing process in bioprocessing and production facilities

1.2 Who Uses Biotechnology?
1.3 The Biotechnology Industry
1.5 Industry Practices
7.6 Protein Production in Industry

Not covered in textbook or activity:

Discuss upstream and downstream processing, including the key steps and equipment involved in each process

Describe environmental monitoring in a controlled space

Understand the importance of audits and corrective and preventive action (CAPA) to ensure compliance and continuous improvement

Discuss attributable, legible, contemporaneous, original, and accurate (ALCOA) principles and FDA guidelines on data integrity and compliance

Discuss strategies for maintaining security of sensitive data and intellectual property

Describe the significance of Health Insurance Portability and Accountability Act (HIPAA) compliance and the protection of personal health information

Biotechnology Skills

Topic	Textbook Content	Lab Activity
Accurately measure liquids using micropipets and serological pipets	2.3 Laboratory Equipment [How To] Use an Adjustable-Volume Micropipet	Activity 2.B Pipetting <i>Numerous other lab activities involve use of pipets and micropipets</i>
Accurately measure mass using electronic balances	2.3 Laboratory Equipment [How To] Use an Adjustable-Volume Micropipet	Activity 2.B Pipetting Activity 2.D Making Solutions Activity 3.A Making Microbiology Media
Properly prepare solutions and buffers	2.5 Preparing Solutions	Activity 2.D Making Solutions Activity 2.E Titration
Properly measure and adjust the pH of a solution with a pH meter	2.3 Laboratory Equipment	Activity 2.D Making Solutions Activity 2.E Titration
Properly perform a serial dilution	3.5 Microbiological Techniques 9.4 Project Ideas	Activity 3.E Quantifying Bacterial Numbers Chapter 3 Extension Activities Chapter 5 Extension Activities Activity 7.E Biofuel Enzyme Assay Activity 8.C Quantitative ELISA
Identify and properly handle reagents	2.1 Laboratory Safety 2.5 Preparing Solutions	Activity 2.D Making Solutions Activity 2.E Titration
Demonstrate proper aseptic/sterile technique	3.5 Microbiological Techniques 3.8 Eukaryotic Cell Culture [How To] Use Aseptic Technique to Transfer Bacteria	Activity 3.A Making Microbiology Media Activity 3.B Disk Diffusion Test (Modified Kirby-Bauer Test) Activity 3.C Microbes and Health: An Illustration of Koch's Postulates Activity 3.D Gram Staining Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 5.C Purification of S3 and pGLO Plasmids Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography
Demonstrate proper culturing of microorganisms	3.4 Culturing Bacteria in the Laboratory 3.5 Microbiological Techniques 3.8 Eukaryotic Cell Culture [How To] Use Aseptic Technique to Transfer Bacteria	Activity 3.A Making Microbiology Media Activity 3.B Disk Diffusion Test (Modified Kirby-Bauer Test) Activity 3.C Microbes and Health: An Illustration of Koch's Postulates Activity 3.D Gram Staining Activity 5.A Bacterial Transformation with S3 Plasmid

		Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 5.C Purification of S3 and pGLO Plasmids Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography
Describe the proper use of a centrifuge	5.3 Plasmid Purification and Quantitation [How To] Use a Microcentrifuge	Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.E Biofuel Enzyme Assay
Describe the proper use of a spectrophotometer	2.3 Laboratory Equipment 3.5 Microbiological Techniques 5.3 Plasmid Purification and Quantitation 7.5 Methods of Protein Analysis 8.3 Immunoassays	Activity 3.F Staining Eukaryotic Cells Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation Activity 7.A Protein Quantitation Using the Bradford Assay Activity 7.E Biofuel Enzyme Assay Activity 8.C Quantitative ELISA
Demonstrate an understanding of how calibration and validation are critical to producing reliable, consistent results	1.5 Industry Practices 2.3 Laboratory Equipment [How To] Use an Adjustable-Volume Micropipet	Activity 2.B Pipetting Activity 2.C Kool-Aid Column Chromatography Activity 2.D Making Solutions Activity 2.E Titration
Describe the proper use and maintenance of biosafety cabinets (BSC)	2.1 Laboratory Safety 2.3 Laboratory Equipment 3.5 Microbiological Techniques	
Demonstrate proper use of electrophoresis equipment	4.4 DNA Analysis Techniques 7.4 Proteins in Biotechnology [How To] Set Up a Vertical Electrophoresis System	Activity 4.C Dye Electrophoresis Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.D DNA Quantitation Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding Activity 7.D SDS-PAGE of Fish Muscle Activity 8.D Western Blotting
Discuss the role and impact of automation in the biotechnology industry	[Careers in Biotech] Timothy Balmorez [Biotech in the Real World] Fighting Crime with DNA 8.3 Immunoassays	
Use 24 hr time correctly	–	–

Applied Mathematics

Topic	Textbook Content	Lab Activity
Use scientific notation correctly	2.4 Numerical Data 3.5 Microbiological Techniques 5.2 Transforming Cells	Activity 3.E Quantifying Bacterial Numbers Activity 5.B Bacterial Transformation with pGLO Plasmid
Use significant digits correctly	2.4 Numerical Data 3.5 Microbiological Techniques 5.2 Transforming Cells	Activity 3.E Quantifying Bacterial Numbers Activity 5.B Bacterial Transformation with pGLO Plasmid
Understand and use mathematical symbols	<i>Not specifically covered but covered in general math and science courses</i>	Activity 3.E Quantifying Bacterial Numbers Activity 5.B Bacterial Transformation with pGLO Plasmid
Understand and use fractions	<i>Not specifically covered but covered in general math and science courses</i>	Activity 5.B Bacterial Transformation with pGLO Plasmid
Use metric measurements and perform metric unit conversions	2.4 Numerical Data 3.5 Microbiological Techniques 5.2 Transforming Cells	Activity 7.A Protein Quantitation Using the Bradford Assay
Generate a graph using collected data	3.5 Microbiological Techniques 5.3 Plasmid Purification and Quantitation 7.5 Methods of Protein Analysis 9.2 Student Research Projects	Activity 3.E Quantifying Bacterial Numbers Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.D DNA Quantitation Activity 7.A Protein Quantitation Using the Bradford Assay Activity 7.E Biofuel Enzyme Assay Activity 8.C Quantitative ELISA
Perform calculations for serial dilutions	3.5 Microbiological Techniques 9.4 Project Ideas	Activity 3.E Quantifying Bacterial Numbers Chapter 3 Extension Activities Chapter 5 Extension Activities Activity 7.E Biofuel Enzyme Assay Activity 8.C Quantitative ELISA
Perform calculations using dilution ratios		
Solve dilution factor calculations	3.5 Microbiological Techniques	Activity 3.E Quantifying Bacterial Numbers Activity 5.D DNA Quantitation Activity 7.E Biofuel Enzyme Assay Activity 8.C Quantitative ELISA
Solve molarity solution calculations	2.5 Preparing Solutions	Activity 2.D Making Solutions
Solve volume/volume (V/V) solution calculations	2.5 Preparing Solutions	Activity 2.D Making Solutions
Solve weight/volume (W/V) solution calculations	2.5 Preparing Solutions	Activity 3.A Making Microbiology Media
Describe the relationship between pH measurements and the logarithmic scale	2.3 Laboratory Equipment	

Apply basic statistical techniques such as mean, median, mode, and standard deviation to analyze data

9.2 Student Research Projects

Activity 2.B Pipetting
Activity 2.E Titration
Activity 8.C Quantitative ELISA

Understand principal component analysis (PCA)

9.2 Student Research Projects

Determine sample size for studies to ensure the reliability and validity of study results

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Standard Equipment

Topic	Textbook Content	Lab Activity
Identify common glassware and equipment	2.3 Laboratory Equipment	<i>Throughout</i>
Demonstrate proper and safe use of:		
▪ Micropipets and serological pipets	2.3 Laboratory Equipment [How To] Use an Adjustable-Volume Micropipet	Activity 2.B Pipetting <i>Numerous other lab activities involve use of pipets and micropipets</i>
▪ Electronic balances	2.3 Laboratory Equipment [How To] Use an Adjustable-Volume Micropipet	Activity 2.B Pipetting Activity 2.D Making Solutions Activity 3.A Making Microbiology Media
▪ pH meters	2.3 Laboratory Equipment	Activity 2.D Making Solutions Activity 2.E Titration
▪ Stirrers/shakers	2.3 Laboratory Equipment 8.3 Immunoassays	Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 8.D Western Blotting
▪ Vortexers		Activity 3.E Quantifying Bacterial Numbers Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion

Water baths and heating/cooling blocks	5.2 Transforming Cells [How To] Set Up a Restriction Digest	Activity 2.A DNA Extraction and Precipitation Activity 3.A Making Microbiology Media Activity 4.B Casting Agarose Gels Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 5.C Purification of S3 and pGLO Plasmids Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.D SDS-PAGE of Fish Muscle Activity 7.E Biofuel Enzyme Assay Activity 8.D Western Blotting
Autoclaves	2.1 Laboratory Safety 2.3 Laboratory Equipment	Activity 2.D Making Solutions Activity 3.A Making Microbiology Media Activity 3.B Disk Diffusion Test (Modified Kirby-Bauer Test)
Biosafety cabinets	2.1 Laboratory Safety 2.3 Laboratory Equipment 3.5 Microbiological Techniques	
Fume hoods	2.1 Laboratory Safety	
Centrifuges	5.3 Plasmid Purification and Quantitation [How To] Use a Microcentrifuge	Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.E Biofuel Enzyme Assay
Spectrophotometers	2.3 Laboratory Equipment 3.5 Microbiological Techniques 5.3 Plasmid Purification and Quantitation 7.5 Methods of Protein Analysis 8.3 Immunoassays	Activity 3.F Staining Eukaryotic Cells Activity 5.C Purification of S3 and pGLO Plasmids Activity 5.D DNA Quantitation Activity 7.A Protein Quantitation Using the Bradford Assay Activity 7.E Biofuel Enzyme Assay Activity 8.C Quantitative ELISA
Microscopes	3.1 Microbiology and Cell Biology 3.2 Bacteria 3.8 Eukaryotic Cell Culture 8.3 Immunoassays	Activity 3.C Microbes and Health: An Illustration of Koch's Postulates Activity 3.D Gram Staining Activity 3.F Staining Eukaryotic Cells
Electrophoresis equipment	4.4 DNA Analysis Techniques 7.4 Proteins in Biotechnology [How To] Set Up a Vertical Electrophoresis System	Activity 4.C Dye Electrophoresis Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.D DNA Quantitation Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding Activity 7.D SDS-PAGE of Fish Muscle Activity 8.D Western Blotting

Incubators	3.5 Microbiological Techniques 3.8 Eukaryotic Cell Culture 5.3 Plasmid Purification and Quantitation	Activity 3.B Disk Diffusion Test (Modified Kirby-Bauer Test) Activity 3.C Microbes and Health: An Illustration of Koch's Postulates Activity 3.E Quantifying Bacterial Numbers Activity 4.D Restriction Digestion and Analysis of Lambda DNA Activity 4.E Forensic DNA Fingerprinting Activity 5.A Bacterial Transformation with S3 Plasmid Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 5.C Purification of S3 and pGLO Plasmids Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography
Chromatography equipment	3.3 Uses of Bacteria in Biotechnology 5.3 Plasmid Purification and Quantitation 7.5 Methods of Protein Analysis 7.6 Protein Production in Industry	Activity 5.C Purification of S3 and pGLO Plasmids Activity 7.B Size Exclusion Chromatography Activity 7.C GFP Purification by Hydrophobic Interaction Chromatography Activity 7.F Exploring Bioinformatics with GFP
Describe proper and safe use of:		
Fourier transform infrared (FTIR) spectrophotometers	—	—
PCR thermal cyclers and real-time PCR thermal cyclers	Chapter 6 The Polymerase Chain Reaction	Activity 6.A STR PCR Analysis Activity 6.B GMO Detection by PCR Activity 6.C Detection of the Human PV92 Alu Insertion Activity 6.D Fish DNA Barcoding
Flow cytometers	8.3 Immunoassays	

Experimental Design and Data Analysis

Topic	Textbook Content	Lab Activity
Read, interpret, and draw conclusions from technical material, test records, and specification sheets	—	—
Discuss good experimental design, including the proper use of controls	4.4 DNA Analysis Techniques Chapter 9 Research Projects [How To] Read a Journal Article 6.2 What Is PCR? 6.4 Types of PCR 6.7 Real-World Applications of PCR [How To] Program a Thermal Cycler 8.3 Immunoassays	Activity 3.B Disk Diffusion Test (Modified Kirby-Bauer Test) Activity 3.C Microbes and Health: An Illustration of Koch's Postulates Activity 4.A Restriction Site Prediction Using NEBcutter Activity 5.B Bacterial Transformation with pGLO Plasmid Activity 6.B GMO Detection by PCR Activity 8.A Ouchterlony Double Immunodiffusion Assay Activity 8.B Serum Antibody Detection by ELISA Activity 8.C Quantitative ELISA

Explain how to maintain a laboratory notebook	2.2 Laboratory Notebooks	
Analyze and interpret data, including the use of statistical analysis	9.2 Student Research Projects	<i>Throughout</i>
Understand and apply the principles of responsible conduct of research (RCR)	–	–
Demonstrate proper implementation of data integrity principles	–	–
Effectively communicate scientific findings	2.4 Numerical Data 9.3 Communicating Research	

Bio-Rad Classroom Teaching Resources

Go to [bio-rad.com/classroomresources](https://www.bio-rad.com/classroomresources) for a range of professional development and teaching resources to support biotechnology and biology education.

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